

Rail Operations Decision Support Dashboard

How can rail operations connect inventory information, anticipate material needs and choose timely actions that improve availability, warehouse planning and resource use?

How to Read This Report

This report explains how the dashboard supports rail operations decisions. Follow the information flow below, then use the contents page to move directly to an area of interest.

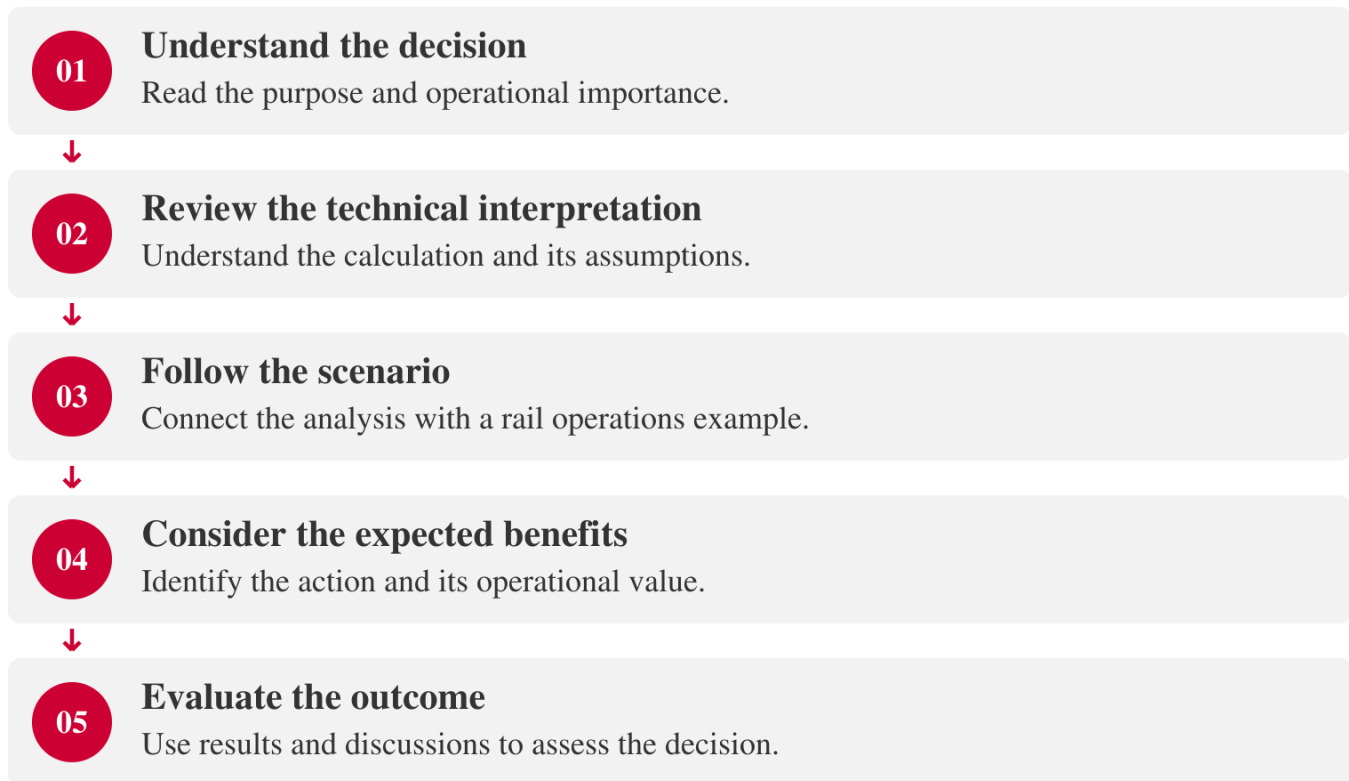


Fig 1: Information flow for reading and applying the report

Reading the Explanations

The narrative explains the purpose and operational importance of each visual. Technical Interpretation describes the calculation and assumptions. Numbered scenarios connect the analysis to practical rail operations situations. Expected Benefits explains how the result can support action.

Interpreting the Evidence

Scenarios are illustrative unless identified as a reported working result. Simulation outputs describe modeled performance rather than guaranteed outcomes. Results and Discussions distinguishes decision-support capability from benefits that must be measured after implementation.

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1. Introduction

Rail operations depend on thousands of material decisions that are individually small but collectively important: whether a part will be available, whether an incoming order can be accommodated, whether an inventory policy still fits demand, and whether a system balance can be trusted. The Warehouse Inventory Intelligence Dashboard brings these questions into a connected decision-support environment.

Its principal value is the ability to move from observation to a defensible action. A quantity becomes more useful when considered alongside consumption and replenishment timing. A recommended stock level becomes more meaningful when its service and cost consequences are visible. A count list becomes more effective when it reflects the consequence of an inaccurate record.

This report explains the three dashboard pages as one operational process. Current Inventory Status identifies where attention is needed. Min–Max Optimization simulates and evaluates multiple inventory policies. Cycle Count Optimization directs physical verification toward the records that matter most under the selected priorities.

2. Background

Rutgers–CAIT developed this work within NJ TRANSIT’s broader PiMS modernization effort. Initial work at the Meadows Maintenance Complex (MMC) connected warehouse space planning with inventory visibility. Site observations, warehouse scanning, location information and inventory extracts provided the basis for relating material records to the physical storage environment. Planning for increased operational demands, including FIFA 2026 preparations, reinforced the need to anticipate incoming material and storage pressure.

The work expanded as the team connected purchasing, usage and inventory information. Questions about where to store material naturally led to questions about which material moves, which parts may run short and which stock deserves review. Simulation-based replenishment analysis and priority-based cycle counting extend that same decision process.

Power BI provides the analytical and reporting environment described in the source. ArcGIS connects inventory records with mapped warehouse regions. The optimization models were built using Python. The website provides a presentation channel for implemented visuals; its behavior and refresh timing depend on the published data and integration in place.

A Connected Decision

A bearing may be easy to overlook in a list of balances. Its importance becomes clear when the user sees frequent usage, a long replacement lead time, a vulnerable inventory position and a mapped storage region together. The intended audience includes operational staff, managers and analytical specialists. The report therefore explains both the practical meaning of a result and the assumptions that govern its interpretation.

3. Problem Statement

NJ TRANSIT’s rail operations require timely, coordinated decisions supported by reliable information. Relevant data is distributed across separate systems, spreadsheets and manual records, making it difficult to establish a shared view of material availability, future requirements and emerging

risk. The source report describes fragmentation across Oracle, Trapeze and local records; combining extracts for analysis does not by itself create a complete operational connection between those systems.

The effect is a substantial interpretation burden on staff. A balance may be available without a dependable view of incoming supply. A location code may exist without intuitive spatial context. An inventory policy may persist even when demand or lead time changes. Teams must reconcile these differences before they can act, often when an operational need is already urgent.

Underlying issue	Operational consequence	What to track
Fragmented information	Time is spent reconciling records and confirming status.	Review turnaround time; reconciliation effort.
Limited forward visibility	Demand or delivery changes can be recognized late.	Material-related delays; emergency purchases.
Static inventory rules	Stock can be excessive for some parts and inadequate for others.	Service availability; average inventory; overdue supply.
Uneven verification priorities	Counting effort may miss records with high operational consequences.	Count hours; discrepancy value; critical record coverage.
Limited spatial evaluation	Record-based decisions may not reflect physical storage constraints.	Search time; location accuracy; usable occupancy.

Table 1: Operational challenges and evaluation measures

Over time, these conditions can reinforce one another. Excess stock can restrict usable storage while shortages of other parts delay planned work. Uncertain records can prompt unnecessary purchases or hide a requirement until the replacement lead time is already too long. Knowledge held in individual emails or personal experience can be difficult to retain and share.

The need is a consistent way to connect evidence, compare choices and assign action before these effects accumulate. The proposed performance measures above are evaluation measures, not claims that their current values have been established.

The Decision Gap

A report can state that stock is low. Decision support must help establish whether it is low for the expected need, whether replenishment is timely, what response is practical and who should verify the next step.

4. Methodology

The methodology follows a decision cycle: establish a reliable inventory picture, interpret the operational exposure, compare a policy or priority, take an approved action, and check the outcome. The three pages share this logic while answering different questions.

Stage	Analytical activity	Decision output
Prepare	Standardize item identifiers, data types, dates and units; reconcile inventory, usage, purchase-order and location records.	A defined, traceable population of items.
Describe	Calculate balances, movement, coverage and relative exposure.	An exception list for operational review.
Compare	Evaluate inventory policies and operating scenarios.	A service–inventory trade-off for approval.
Prioritize	Rank count exposure and select items against coverage and mandatory rules.	A feasible physical verification plan.
Learn	Compare physical counts and actual service with assumptions.	Corrected records and revised planning inputs.

Table 2: Methodology from data preparation to decision review

Data Preparation Determines Whether The Visual Is Trustworthy

Item identifiers must match across sources. Usage quantities require a consistent sign convention and observation period. Dates must distinguish a historical procurement lead time from an actual promised delivery date. Inventory held at multiple locations must be aggregated deliberately. If one item is joined to several purchase-order lines, repeating its on-hand balance can inflate both quantities and value unless the item-level calculation is protected.

What Users See Versus Where Calculations Happen

Power BI measures summarize and filter the prepared data. Optimization and simulation results can be calculated upstream and loaded into the reporting model. Selecting a scenario may therefore select an existing result rather than run a new simulation. Updated assumptions require the appropriate calculation and refresh process; an email or a website click does not automatically update the model.

Evidence Boundaries

- A source-described capability is explained as documented, without implying an independent audit of the deployed code.
- Project working results and scenario examples are labeled separately from observed operational outcomes.
- Missing usage, prices, lead times or location links should be visible as data exceptions rather than interpreted as zero.
- An operational user validates exceptions and approves changes in the relevant system of record.

Technical Interpretation

Formula notes explain the intended relationship. Exact thresholds, scenario definitions, weighting variants and validation rules belong to the versioned implementation and must remain consistent with the data published to users.

4.1 Current Inventory Status

4.1.1 Summary Scorecards and Filters

The first page establishes the scale of the inventory population under review. Summary scorecards show measures such as quantity on hand, open purchase-order quantity, estimated inventory value and the number of tracked items or records. Together they answer a practical opening question: how much material is held, how much is coming, and how large is the financial exposure?

The scorecards are useful as a shared starting point, not as a complete health assessment. A large quantity can conceal a shortage of one essential part, and a large incoming order does not confirm that storage is available. Users should move from the totals to the visuals that explain which items drive them.

Technical Interpretation

On-hand quantity is aggregated over the intended item and location population. Open-order quantity is aggregated over qualifying order lines. Estimated inventory value is the sum of each item's eligible quantity multiplied by its unit price. Distinct item count and record count answer different questions and should be labeled accordingly. Prices, units and inclusion rules must be consistent before aggregation.

Filters and supported selections narrow the review to an item group or mapped area. Users must check which filters apply to each visual: the source describes interactive Power BI filtering, but that does not establish identical behavior for every website scorecard.

Scenario 1

Before an incoming shipment, the manager reviews the relevant pallet area. A high total quantity on order prompts a closer look at the arriving parts. Large wheel assemblies and small electrical items may have similar unit totals but very different storage requirements. The scorecards identify the scale of the review; the item and spatial views provide the context for the decision.

Expected Benefits

- Start operational reviews from the same inventory scope and refresh date.
- Identify whether the main concern is inventory investment, incoming volume or a specific item exception.
- Use the selected population to focus subsequent reviews rather than treating all stock equally.

Interpretation Limits

Adding quantities across different parts does not measure physical space or service readiness. Inventory value is a stock valuation measure, not an available saving. Missing prices should reduce

reported valuation coverage rather than silently imply zero value. A timestamp shows when data was exported or refreshed; it does not guarantee that every source transaction is current.

The manager's next question is therefore not simply "Is the total high?" but "Which parts within this total need a different operational response?"

4.1.2 High-Movement Inventory and Slotting Priority

Movement analysis identifies parts consumed frequently relative to available stock. Slotting priority uses this context to help users recognize material that may deserve easier access, closer monitoring or more deliberate placement. Its value is greatest when movement is read alongside supply pressure: a fast-moving part with reliable local replenishment is different from one with little available stock and a long replacement lead time.

This visual supports a review of handling priorities. It does not, by itself, solve a physical placement problem. A part can be important to retrieve quickly while still requiring a particular rack, handling method or storage restriction.

Technical Interpretation

The source describes movement relative to stock, with inventory gap and lead time as additional context. A usage-to-stock relationship can highlight how intensely stock is being consumed, provided the usage period is consistent. Zero stock requires explicit handling; missing usage cannot be assumed to mean no demand. The exact slotting score and category thresholds are not specified in the supplied report.

Scenario 2

Two filter kits are issued regularly. One is stocked in depth and can be replenished locally in five days. The other has limited stock and a 60-day lead time. Both may deserve convenient access, but the second also needs an immediate supply review. The visual helps distinguish a handling improvement from a replenishment concern.

4.1.3 Inventory Gap as Supporting Context

The source defines inventory gap as quantity ordered minus quantity on hand. This is a comparison between incoming and existing quantities. It must not be interpreted as unmet demand: it contains no requirement quantity and does not establish when ordered stock will arrive.

Technical Interpretation

Source relationship: $\text{inventory gap} = \text{quantity ordered} - \text{quantity on hand}$. A positive result indicates that the ordered quantity exceeds the current on-hand balance. A shortage assessment additionally requires demand and replenishment timing.

Expected Benefits

- Review access and handling for parts with sustained movement.
- Separate frequent usage from genuine supply vulnerability.
- Check capacity, dimensions, load limits and operational restrictions before relocating material.

The expected benefit is a better reason for choosing which parts to review first. Reduced walking time, improved retrieval speed and fewer handling interruptions should be measured during an operational pilot rather than assumed from the ranking.

4.1.4 Inventory Coverage and Service Availability

Inventory coverage converts a balance into an estimate of how long it can support expected usage. Comparing that duration with replenishment timing makes the visual useful for anticipating shortages. Ten units can represent several months of protection for one part and only a few days for another.

Users should read the visual as an early-warning screen. An exposed item triggers a check of actual requirements, usable stock, open orders and delivery commitments. That review may lead to expediting, transferring material, reserving stock for critical work or adjusting a schedule.

Technical Interpretation

Estimated daily usage equals historical usage divided by the relevant observation period. Days of inventory coverage equals usable quantity on hand divided by estimated daily usage. The two periods must use compatible units. If a confirmed receipt date is available, compare coverage with the days remaining until that receipt. A typical historical lead time is a planning proxy, not a confirmed arrival date.

Scenario 3

A relay has six usable units available and expected demand of two units per week. It provides about three weeks of coverage. If the next usable receipt is six weeks away, expected demand before receipt is 12 units, leaving a potential gap of six. The manager can investigate a transfer or earlier partial shipment before the relay is requested for planned work.

Question	What the user checks
Is the balance usable?	Exclude or separately identify reserved, damaged or unavailable stock where the data supports it.
Is the demand estimate appropriate?	Check an upcoming campaign, intermittent usage or known one-time requirement.
Will supply arrive in time?	Validate promised dates, remaining order quantities and receipt availability.

Table 3: Checks supporting an inventory coverage decision

Expected Benefits

The visual creates response time. Its operational value is the interval between recognizing the exposure and the date material is needed. Earlier recognition can improve the chance of finding a practical response without an emergency purchase or a last-minute schedule change.

Technical Interpretation

A simple coverage ratio assumes a representative usage rate. It does not fully model demand spikes, multiple receipts or uncertain delivery dates. Zero historical usage makes the ratio undefined or unsuitable, not proof that a part is safe indefinitely. Min–Max scenario analysis extends the review by examining uncertainty rather than relying solely on the average rate.

4.1.5 Slow-Moving, Dormant and Obsolete-Review Inventory

Movement and inactivity views identify stock that deserves a closer review of purpose, retention and location. Long inactivity can signal a changed equipment requirement, excess holdings or an item that no longer supports the active fleet. It can also describe an intentionally retained emergency spare.

Technical Interpretation

Movement categories use usage history and the last-used date. The project’s stated inactivity review threshold is 730 days; the source report does not reproduce the full category logic. A threshold crossing should initiate a review under the applicable retention rules. Missing dates must be distinguished from confirmed inactivity, and any dormant-stock share must identify its denominator: items, units or value.

Scenario 4

A large assembly has not been issued for more than two years. Before freeing its storage position, the team checks fleet compatibility, approved substitutes and protected-spare status. If it must remain, it may still be suitable for a less accessible storage area. The decision is whether and where to retain it, not automatic disposal.

4.1.6 Procurement Exposure

Procurement exposure highlights parts whose acquisition characteristics warrant attention. High price, large order quantity and long lead time can combine to create a difficult replenishment decision. Reading this alongside movement and coverage helps users distinguish routine follow-up from an item requiring early coordination.

Technical Interpretation

The project’s working exposure relationship is $\text{unit price} \times \text{purchase-order quantity} \times \text{lead time}$. If lead time is in days, the raw product has value-days units. It is a relative prioritization measure, not a dollar amount, expected loss or probability of supplier failure. The source also discusses inventory pressure as interpretive context; its inclusion in the implemented score must follow the documented formula.

Scenario 5

An imported wheel costs \$4,800 and is purchased in a batch of 24 with a 120-day lead time. The order value is \$115,200. The raw exposure product is 13,824,000 value-days. That large score points to a need for early delivery follow-up; it does not mean a \$13.8 million loss.

Expected Benefits

Together, these views help distinguish stock that needs retention review from supply that needs protection. Teams can investigate storage opportunities while maintaining attention on difficult-to-replace material. Actual space release and avoided cost depend on approved actions, not on the score alone.

4.1.7 Spatial View and Item-Level Recommendations

A risk signal becomes actionable when staff can connect it to a physical storage area and an identifiable item. ArcGIS provides this spatial context by representing mapped warehouse regions as selectable polygons. Supported region selections filter related inventory information in Power BI.

The item-level recommendation view completes the review by bringing the part identifier, movement, supply exposure and location context together. Where the website presents a simplified set of visuals, detailed investigation may continue in Power BI or the relevant operational records. The report does not assume every Power BI interaction is reproduced on the website.

Technical Interpretation

Warehouse scan regions are cleaned and assigned a stable mapping key. Inventory location fields are linked to the corresponding region through that key or a maintained location mapping. A selection changes the associated inventory filter context. Unmatched locations should be reported; otherwise the visible map can appear more complete than the inventory coverage actually is.

Scenario 6

The system shows a bearing in stock, but an employee cannot find it at the expected position. The manager uses the mapped region and item information to organize a location check before treating the problem as a stockout. If the bearing is found elsewhere, the response is a location correction; if it is absent, a quantity correction and supply review may be required.

Expected Benefits

The map helps bridge the gap between an analytical exception and physical work. It can support a targeted location review, preparation for an incoming delivery or investigation of stock that is slow to retrieve. It should not be interpreted as live tracking of every part.

Supported interpretation	Further data needed
Understand the mapped region associated with an item.	Complete and maintained item-to-location links.
Investigate access and regional storage pressure.	Current occupancy and usable capacity.
Develop future exact placement recommendations.	Part dimensions, weights, package quantities, load limits and storage restrictions.

Table 4: Spatial capabilities and additional data requirements

The source states that spatial integration is primarily at storage-category or region level. Exact aisle/bin placement, complete Boneyard coverage and capacity-constrained slotting are development opportunities rather than established outcomes.

Decision Handoff

End the current-status review with an item, a reason for attention, a proposed response and an owner. For example: verify bearing quantity in the mapped area, confirm the incoming delivery, then review whether the replenishment policy remains appropriate.

4.2 Min–Max Optimization

4.2.1 Policy and Operating-Scenario Selectors

Min–Max planning asks when replenishment should start and how far inventory should be restored. The Min acts as the reorder trigger under the implemented policy; the Max defines the target level used when ordering. A useful recommendation must account for demand, replenishment time and the desired level of service.

The selectors let the manager compare Lean, Balanced and High Service policies under available demand and lead-time scenarios. Stable, moderate and spiky demand represent different operating patterns. Lead-time scenarios represent supply delay assumptions. These controls make the consequences of uncertainty visible before a policy is approved.

Technical Interpretation

The project uses repeated inventory simulations informed by historical usage and lead time. A Poisson-based demand model was used in the project’s modeling work; more variable demand scenarios stress the baseline assumptions. Candidate policies are assessed through measures such as fill rate, average on-hand stock, stockout days and order frequency. The selected result is an outcome of the modeled conditions, not a guaranteed future performance.

Scenario selectors may display precomputed combinations. A policy can be tested under several demand and delay conditions without its recommended Min and Max changing for each selection. Users should distinguish “performance of this policy under stress” from “a newly optimized policy for these assumptions.”

Scenario 7

Thirty imported wheels are available. Normal demand of six wheels per month consumes about 24 over a 120-day lead time, using a 30-day planning month. If demand rises to nine per month and delivery stretches to 156 days, expected consumption becomes about 47 wheels. The potential pre-arrival shortage is about 17. An incoming shipment does not protect the intervening period if it arrives after stock is exhausted.

What Changes The Decision

The manager can compare earlier partial delivery, stock transfer, prioritization of work and a more protective policy. An additional 24-wheel batch at \$4,800 per wheel commits \$115,200, but placing the order only helps the immediate shortage if its delivery can occur in time.

- Test combined demand and supply pressure rather than reviewing either in isolation.
- Compare the investment required for stronger protection.
- Use known future requirements to challenge historical assumptions, then rerun the model where necessary.

The expected benefit is more time to choose a feasible response and a clearer explanation for why an inventory policy should change.

4.2.2 Summary Scorecards

The top-level measures summarize the optimization run. Parts evaluated describes the modeling population. Validated recommendations indicates how many candidates satisfied the implemented model checks. Net inventory value reduction summarizes the modeled financial difference. Replenishment workload multiple indicates the relative ordering burden under the comparison used by the model.

These measures answer different management questions. A large financial opportunity may require more frequent ordering. A high validation count does not mean every item is ready for implementation. Business review still needs to account for campaign demand, protected spares, supplier constraints and unusual operating conditions.

Technical Interpretation

A portfolio inventory-value comparison should aggregate comparable current-policy and recommended-policy simulated average holdings for the same eligible items and price basis. It must not mix a single-day on-hand snapshot with a long-run simulated average without clearly explaining the difference. Items without adequate data should be identified separately.

Technical Interpretation

Net modeled inventory value reduction = comparable current-policy average inventory value – recommended-policy average inventory value. A workload multiple of 2.0 would mean twice the reference ordering workload, not a 2% increase. The actual numerator, denominator and aggregation should be documented with the run.

Scenario 8

A portfolio review suggests that average inventory could fall from \$10.0 million to \$4.8 million while meeting the model's service checks. The \$5.2 million difference is a planning opportunity. Management first checks which items drive it, whether order frequency is manageable, and whether stock can be drawn down through normal usage. It is not immediately available cash.

From A Scorecard To An Implementation Decision

- Review validated candidates through an operational approval process.
- Separate reductions from items that require additional inventory.
- Check purchasing capacity, minimum order quantities and supplier packaging constraints.
- Track realized inventory change and service performance after approved implementation.

The project describes these top metrics as optimization-run outputs rather than measures that necessarily respond to every scenario slicer. The reporting interface should make that scope clear so viewers do not mistake a portfolio result for the currently selected item or stress condition.

The supplied reference does not present a numerical optimization-run summary. This report therefore explains the measures without presenting unverified portfolio values as observed results.

4.2.3 Inventory-Value and Fill-Rate Distributions

An average can hide a wide range of outcomes. Distribution visuals show how results are spread under the selected comparison, helping managers see whether performance is concentrated or variable. The inventory-value view describes the investment associated with a policy. The fill-rate view describes the share of demand fulfilled under the modeled definition.

The chart's unit of observation matters. A distribution across items answers how consistently the policy supports the portfolio. A distribution across simulation runs answers how variable one policy's possible performance is. These interpretations are different; the chart labels and supporting data should establish which is being displayed.

Technical Interpretation

Simulation outputs or item-level results are grouped into the implemented ranges and summarized for each policy. Percentage values must use a consistent scale. Aggregating fill rates also requires care: an unweighted average across parts can differ from total units supplied divided by total units demanded. The source does not specify the distribution bins or aggregation method.

Scenario 9

A cooling fan initially appears suitable for a lower-inventory policy. Before approval, a maintenance forecast raises expected demand from three to five units per month, while a supplier update extends lead time from 150 to 180 days. The team revises the assumptions and reviews new results before accepting the inventory reduction.

Policy	Min–Max	Average inventory value	Expected fill rate
Current	20–60	\$112,000	99%
Lean	15–30	\$57,600	95%
Balanced	20–38	\$76,800	97%
High Service	28–48	\$102,400	99%

Table 5: Illustrative inventory policy comparison

Illustrative scenario values only. They are not reproduced simulation outputs or a measured saving. The table is a companion comparison; it does not reconstruct an unknown distribution.

Expected Benefits

The initial Balanced comparison suggests \$35,200 less average inventory value than the current policy. New information may justify retaining more protection during the campaign. The visuals help

managers explain why a financially attractive reduction should be modified, delayed or limited to selected items.

The expected benefit is a transparent trade-off: how much investment is associated with a service level, how variable that result is, and whether it remains acceptable under the circumstances being planned for.

4.2.4 Current and Recommended Min–Max Table

The item-level table translates the policy comparison into settings that can be reviewed. It should allow users to distinguish current limits, the selected recommendation and associated service or inventory measures. The most useful review asks what would change in practice: a smaller order, earlier replenishment, additional protection or no immediate change.

Technical Interpretation

In a simple order-up-to example, order quantity equals Max minus inventory position when the reorder condition is met. Inventory position usually considers usable on hand, eligible open supply and outstanding demand under the implemented policy. An on-hand-only example is valid only when those other terms are absent or deliberately excluded. Minimum order quantities and pack sizes can change the final purchase quantity.

Scenario 10

An HVAC filter has 28 units on hand, with no open supply or outstanding demand in this simplified example. Current limits of 50–200 imply an order of 172 units; proposed limits of 35–90 imply 62. The difference is 110 units, or \$16,500 at \$150 each. Before changing the settings, the team confirms that upcoming work does not require the additional material.

4.2.5 Stockout-Risk Distribution

The risk-band visual shows how items are distributed across defined ranges of modeled stockout-related exposure. It helps users identify the portion of the portfolio requiring closer attention and, where supported, select a band to review its items.

The source defines service risk as 1 minus the minimum P05 fill rate. The P05 is the lower fifth-percentile fill-rate result; taking the minimum identifies the least favorable included scenario. This creates a conservative service-shortfall score. It is not the probability that an item will experience any stockout. Band boundaries and the scope of the minimum must be documented.

Scenario 11

Before a maintenance campaign, a higher-risk band contains a fuse with four usable units, demand of two per week and a six-week replenishment time. The manager verifies the potential eight-unit pre-arrival shortfall and investigates earlier supply. The band focuses attention; the item review determines the response.

Expected Benefits

The table provides the proposed policy action; the bands organize the exception review. Together they support controlled implementation and monitoring. A band label alone should not determine an

order, and the source does not establish a live automatic link from Min–Max scenario selection to every cycle-count ranking.

4.3 Cycle Count Optimization

4.3.1 Priority Scenarios, Coverage Target and Scorecards

Cycle counting checks whether physical stock agrees with the system record. The dashboard prioritizes the consequence of an error rather than claiming to predict which record is wrong. This distinction makes the approach useful even when historical discrepancy outcomes are unavailable.

Five priority scenarios provide different emphases for the same population. The project discussions describe overall exposure, inventory value, usage, lead time and service risk perspectives; final control labels and weighting definitions should match the implementation. A manager chooses the perspective that fits the operating situation, then adjusts the coverage target while checking the selected workload.

Technical Interpretation

The source identifies inventory value, annual usage, lead time and service risk as priority factors. Factors must be transformed onto comparable scales before combination. CRITIC-family weighting uses the information supplied by each factor, considering variation and overlap between factors. The source identifies D-CRITIC as the preferred production direction; it does not provide sufficient formulas or comparative test results to independently verify that choice.

Ranking and selection are separate steps. The selection layer aims to reach the requested share of total priority exposure with a small practical item set, subject to implemented mandatory and location rules. With equal item effort and no additional constraints, a descending-score selection is a useful benchmark. Different count durations require a workload-aware formulation.

Scenario 12

Two employees can verify about 45 items during the available assignment. The manager starts from the reported working baseline of 45 items covering 61.28% of priority exposure. If a supplier disruption becomes the dominant concern, the lead-time perspective is reviewed. The resulting count requirement must be recalculated; the baseline percentage does not automatically transfer to the new scenario.

What The Controls Protect

- Priority scenario: determines the relative importance of the factors.
- Coverage target: determines how much modeled priority exposure to include.
- Selected-items scorecard: shows the resulting item workload.
- Mandatory-items scorecard: makes required inclusions visible.

Mandatory items are subject to configured business rules, not assumed from the part name. If mandatory work already exceeds capacity, lowering the target cannot resolve the conflict. The manager must escalate staffing or scheduling while preserving required checks.

4.3.2 Cumulative Exposure Curve and Knee Point

The cumulative curve orders items by priority and shows how much total priority exposure is covered as more items are selected. A steep early rise means exposure is concentrated in the leading items. A flatter curve means it is more widely distributed. The knee marks a modeled point where additional counting begins to add less exposure coverage.

The knee is a planning recommendation, not a regulatory stopping rule or proof that all remaining items can be ignored. Count obligations, staff availability and known exceptions remain relevant. The curve helps managers discuss the marginal benefit of additional work using a common basis.

Technical Interpretation

For priority scores s , cumulative coverage at rank k is the sum of scores through rank k divided by the total score. Scores must be comparable and the denominator nonzero. The knee-detection rule selects a cutoff from the resulting curve. The exact algorithm and whether mandatory inclusions are applied before or after detection should be documented; the source does not specify them.

Reported project working baseline	Meaning
45 selected items from 105 eligible items	About 42.9% of the item population is selected.
61.28% of total priority exposure	Coverage of the modeled score, not inventory accuracy or inventory value.
60 items outside the initial selection	These remain subject to the continuing count program.

Table 6: Reported cycle-count baseline and interpretation

Scenario 13

A port disruption affects imported components. After validated input updates and selection of the lead-time perspective, a demonstration result shifts the knee from 45 to 63 items. This would indicate a broader verification requirement under that scoring context. The value 63 is illustrative: the model could move the knee in either direction, or leave it unchanged.

What Management Can Decide

If exposure spreads, temporarily extend the count scope or add staff. If it concentrates, verify whether the leading items merit immediate attention while the rest remain on schedule. Always examine which items changed, not just the cutoff number.

Switching weights changes what “total exposure” means. Coverage percentages from different scenarios are not directly comparable measures of the same underlying risk. To distinguish changed operating conditions from changed preferences, compare updated data under fixed weights as well as comparing alternative scenarios.

Expected Benefits

The curve makes the additional coverage from more counting visible. It supports a reasoned workload discussion without claiming that selected records are more likely to contain errors.

4.3.3 Item Distribution by Scenario

A selected-item total can stay unchanged while the count list changes substantially. The item distribution view helps users understand that change. Its operational purpose is to distinguish items retained from the baseline, items deferred and items newly included under another priority perspective.

A distribution chart may summarize scenario selections without directly displaying set overlap. In that case, the retained, removed and added groups must be obtained by comparing item identifiers in the supporting scenario-item results. The visual should not imply that a total alone establishes which items changed.

Technical Interpretation

Let the baseline selected set be A and the alternative set be B. Retained items are in both A and B; deferred items are only in A; newly included items are only in B. Compare distinct item keys and a consistent eligible population. Mandatory items should remain included when required by the same rules.

Illustrative selection change	Items	Operational interpretation
Retained	34	Important under both priorities.
Deferred from immediate count	11	Remain in the ongoing count program.
Newly included	11	Receive attention under the revised priority.
Final selected workload	45	34 retained + 11 newly included.

Table 7: Illustrative changes in the selected count list

Scenario 14

A local filter kit with a five-day replenishment time is deferred, while an imported bearing with a 120-day lead time enters the immediate list. The count finds three bearings where the system shows six. Staff investigate and correct the discrepancy through the approved process, while the manager checks transfers or delivery options before scheduled work is affected.

Expected Benefits

The same number of checks can be directed toward a more relevant set of records. The practical value is not the substitution itself; it is whether the revised list improves the timing and usefulness of the discrepancies discovered. “Same 45 counts” also does not necessarily mean identical labor: location, packaging and handling requirements can change the time needed per item.

- Issue a dated list identifying the selected scenario and required items.
- Retain deferred items in the normal schedule rather than removing them indefinitely.
- Record physical quantity, discrepancy, reason and time spent.
- Use the outcomes to evaluate future prioritization and data corrections.

Baseline and lead-time selections can be compared at a common workload for this scenario. That is a different management choice from adopting the alternative scenario's knee-point count, which may require more or fewer than 45 items.

5. Results and Discussions

The Result Is A Better-Supported Decision

The reference demonstrates a connected decision-support design, but it does not provide a controlled before-and-after assessment of service, cost or labor. The strongest current conclusion is that the dashboard makes relationships and trade-offs visible in one review process. Operational gains must be confirmed after users act on those results.

Decision capability	Evidence presented	Value and limit
Identify emerging shortages	Source-described coverage analysis; relay and wheel illustrations.	Supports earlier intervention; actual warning accuracy needs observed outcomes.
Compare replenishment policies	Source-described scenario analysis; illustrative filter and fan comparisons.	Makes inventory/service trade-offs explicit; savings depend on approved execution.
Focus verification	Reported working baseline: 45 items, 61.28% exposure.	Concentrates modeled exposure; does not establish discrepancy prediction accuracy.
Connect records to locations	Source-described ArcGIS region filtering.	Supports targeted investigation; does not establish exact bin placement.
Review dormant stock	Source-described inactivity screening.	Supports retention and space review; does not authorize disposal.

Table 8: Decision-support capabilities and evidence

What The Cycle-Count Baseline Demonstrates

Selecting 45 of 105 items represents 42.9% of the item population while covering 61.28% of the modeled exposure. The average exposure per selected item is approximately 2.11 times that of the remaining 60 items: $(61.28 \div 45) \div (38.72 \div 60)$. This demonstrates concentration within the model, not twice the likelihood of an error or twice the realized business benefit.

If every item took the same 12 minutes to count, the first-wave effort would be nine labor-hours versus 21 for all 105 items. The 12-hour difference is an illustrative initial workload reduction. It is not an annual saving because deferred items may still need counting, and real count times vary.

What The Financial Examples Demonstrate

The filter example links a parameter change to 110 fewer units in one simplified replenishment order. The cooling-fan example links a policy choice to a modeled average investment difference. These

show how to reason about decisions. They should not be added together or reported as measured savings.

The Main Selling Point

The dashboard makes it possible to explain an action: what changed, which items are affected, what trade-off is involved and what must be checked before implementation. This is the bridge from information visibility to operational decision-making.

Close The Loop Between Insight And Operational Value

Using The Results During Daily Operations

A useful review begins with data freshness and a defined scope. The user identifies an exception, tests the assumptions behind it, compares a feasible response and records an owner. After the action, the team checks whether the intended outcome occurred. The dashboard supports this cycle; completion still depends on operational coordination and accurate source records.

For example, a shortage signal can lead to an expedited delivery, a transfer, a revised work sequence or an inventory correction. The appropriate response depends on timing, cost and availability. The most valuable output is therefore a justified choice, not simply a higher or lower score.

Expected benefit	Practical measurement	Evidence to retain
Earlier supply intervention	Time from warning to material need; share of warnings confirmed.	Dated warning, validated need and final receipt.
More reliable availability	Demand fulfilled on time; material-related work delays.	Issues, unfilled requirements and work records.
Better inventory investment	Average inventory value alongside service outcomes.	Comparable periods, item scope and policy-change dates.
More effective counting	Count hours, discrepancy value and exposure verified.	Physical count results, item selection and time logs.
Better storage use	Usable occupancy and retrieval time after changes.	Location checks, capacity constraints and before/after observations.
Faster coordinated decisions	Time from exception to assigned and completed action.	Decision log, owner and completion evidence.

Table 9: Operational benefits and measures for evaluation

Fresh Information, Not An Assumption Of Live Operation

The dashboard can support decisions during the working day, but responsiveness depends on extraction, calculation, publication and source-entry timing. A website snapshot is not a live transaction feed. The report should be read alongside the displayed refresh timestamp and known data delays, especially before acting on a critical balance or promised date.

Development Priorities

Near-term work should improve missing-data visibility, timestamp clarity, model documentation and outcome collection. Maintenance requirements and supplier commitments can strengthen forward planning. Dimensions, package quantities, capacity and load limits are prerequisites for exact placement. Historical count discrepancies can later support testing whether predictive error models add value beyond consequence-based ranking.

Advanced forecasting, approved workflow automation and expansion to other facilities are credible next steps when the required data and governance are available. They are not established benefits of the current visuals alone.

6. Conclusion

The Warehouse Inventory Intelligence Dashboard provides a structured way to connect material information with decisions across rail operations. Its three pages form a coherent sequence: understand current conditions, compare inventory policies, and verify the records with the greatest operational consequence.

Current Inventory Status turns balances, usage, incoming supply and location information into a focused review of material pressure and storage priorities. Min–Max Optimization makes the service and investment consequences of policy choices visible under different assumptions. Cycle Count Optimization translates a priority perspective and coverage objective into a practical verification plan.

The principal value is a more consistent basis for action. Users can identify what deserves attention, explain why it matters, compare alternatives and coordinate a response before an issue becomes urgent. Technical calculations remain available for analytical review while the operational questions remain accessible to users without modeling expertise.

The next measure of success is whether decisions improve in practice. A disciplined pilot should connect recommendations to approved actions and observed results in service, inventory, counting effort and storage use. This will establish where the tool delivers value today and where additional data or capabilities are needed.